

- 4 Mexican spadefoot toads, *Spea multiplicata*, live on land but return to ponds to breed. Eggs are laid in water and hatch into tadpoles, which feed in ponds before developing into adults.

The tadpoles can be classified into two main types: omnivore-type tadpoles and carnivore-type tadpoles. Differences between the phenotypes of these two types of tadpole are related to their different feeding behaviours.

- Omnivore-type tadpoles feed on tiny pieces of detritus (dead material from plants and animals) and algae (microscopic photosynthetic organisms) at the bottom of ponds. These tadpoles grow slowly.
- Carnivore-type tadpoles feed on small animals in the water, such as fairy shrimp and small omnivore tadpoles. These tadpoles grow quickly.

Fig. 4.1 shows two tadpoles of the same age, one of each type. A fairy shrimp is also shown. All three organisms are at the same distance from the camera.



Fig. 4.1

Between these two main types of tadpole there is a continuous range of tadpoles with intermediate body phenotypes and feeding behaviours.

For any individual tadpole, regardless of age, it is possible to calculate a phenotype score depending on the features of the tadpole. A tadpole with a phenotype score close to 3 is a typical omnivore type and a tadpole with a phenotype score close to 7 is a typical carnivore type.

The phenotype scores were determined for a large number of tadpoles sampled from two ponds. The availability of detritus and algae was high for one pond and low for the other pond. All other conditions in the two ponds were similar.

For both ponds, phenotype scores were determined shortly after the tadpoles had hatched from eggs and ten days later.

The results are shown in Fig. 4.2.

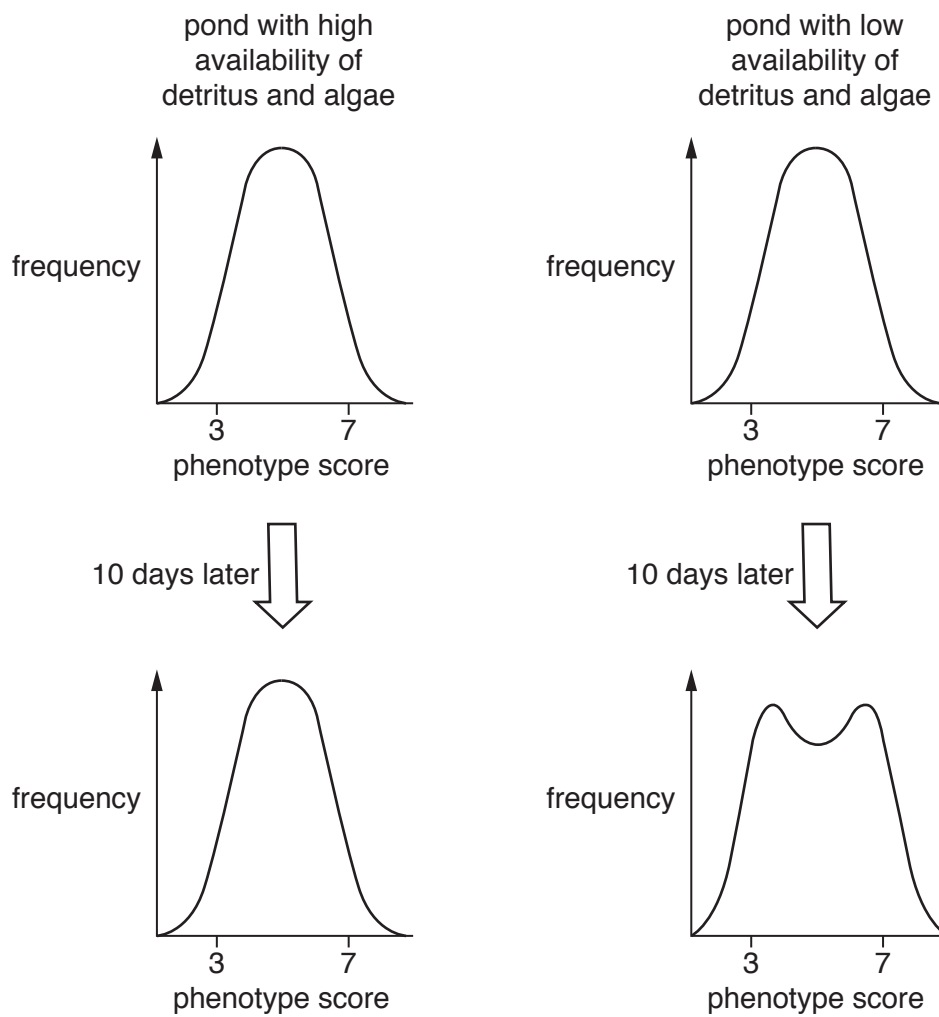


Fig. 4.2

- (a) Using the results shown in Fig. 4.2, describe **and** suggest an explanation for the change in frequency of tadpole phenotypes in the pond with **low** availability of detritus and algae.

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- (b) Eleven days after eggs had hatched, the mean body masses of omnivore-type tadpoles and carnivore-type tadpoles in the pond with low availability of detritus and algae were measured. The results are shown in Fig. 4.3.

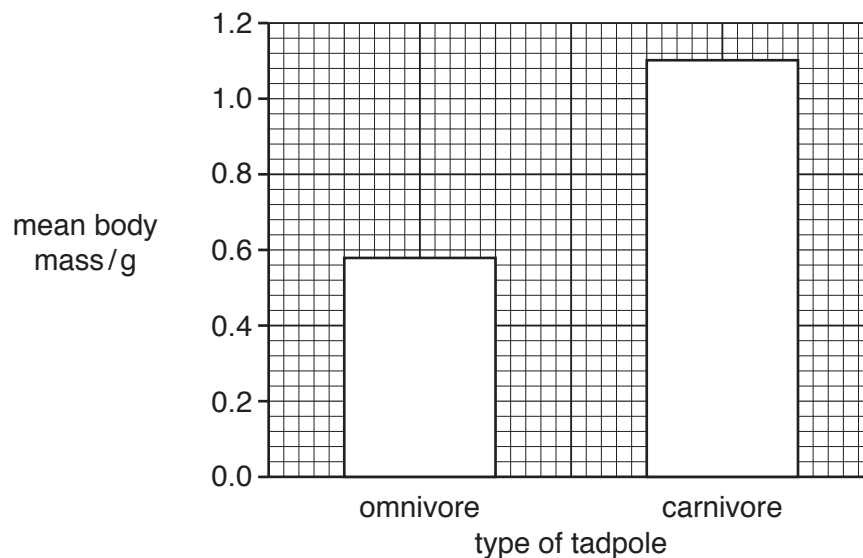


Fig. 4.3

Using the data in Fig. 4.3, calculate the mean body mass of an omnivore-type tadpole as a percentage of the mean body mass of a carnivore-type tadpole.

Show your working.

percentage =%

[2]

- (c) The phenotype scores of a large number of tadpoles were determined in a different pond with **very low** availability of detritus and algae. All other conditions were the same as the first two ponds. As previously, measurements were taken shortly after the eggs hatched and ten days later. After ten days, nearly all of the Mexican spadefoot toad tadpoles in this pond were carnivore types.

The results are shown in Fig. 4.4.

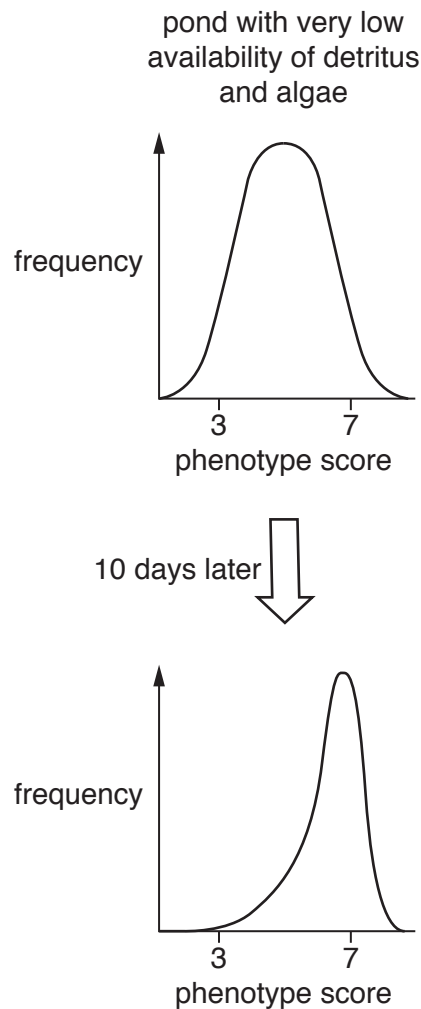


Fig. 4.4

- (i) State the type of natural selection that is acting on the tadpoles in the pond with **very low** availability of detritus and algae.

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- (ii) Suggest explanations for the change in phenotype frequencies of the tadpoles in the pond with **very low** availability of detritus and algae, as shown in Fig. 4.4.

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(d) Mexican spadefoot toad tadpoles develop into adult toads that do not live in water.

In some years, the ponds where Mexican spadefoot toad tadpoles live, dry out quickly.

Suggest why the carnivore-type tadpoles have a selective advantage in the years when ponds dry out quickly.

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[Total: 10]

Question 5 starts on page 20